sector_balanced_max

equity strategy | family: equity-max | status: research | derived from `sector_balanced`

What this strategy does

sector_balanced with bigger bets for more return: base unit 0.012, 6x leverage cap

Why it should work. Same drift, with sector and size risk diversified away so only the earnings signal is left.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.012
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

metric	sector balanced max	market (CRSP VW)
Annualized return	8.35%	9.94%
Annualized volatility	12.82%	19.35%
Sharpe (NAV-pct)	0.69	0.59
Sortino	0.98	
Max drawdown	-44.9%	-55.5%
Calmar	0.19	
Final NAV ($10M start)	$47,669,815	
Worst day	-7.42%	
% positive days	49.3%	
Sharpe 1996-2001	1.20	

Sharpe 2002-2007	1.39	
Sharpe 2008-2013	-0.41	

Correlation of daily returns with the market: **-0.02**.

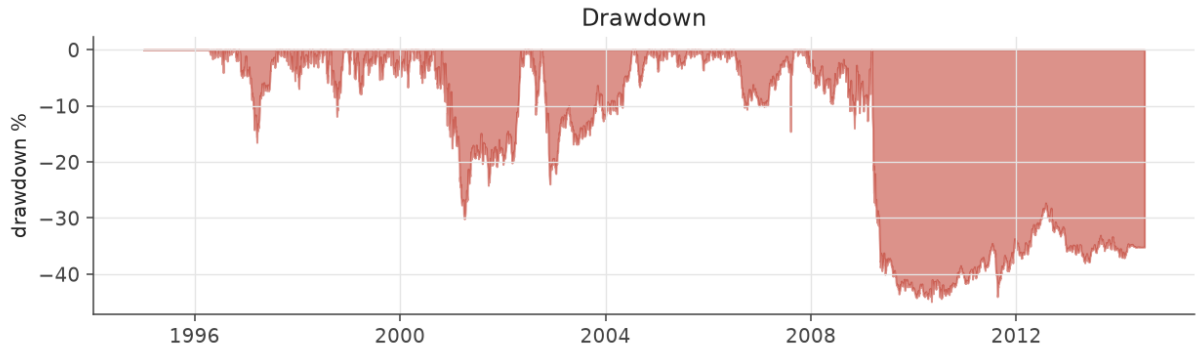
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 0.69

• best return among equity: `market_boost_max` (0.15); this one: 0.0835

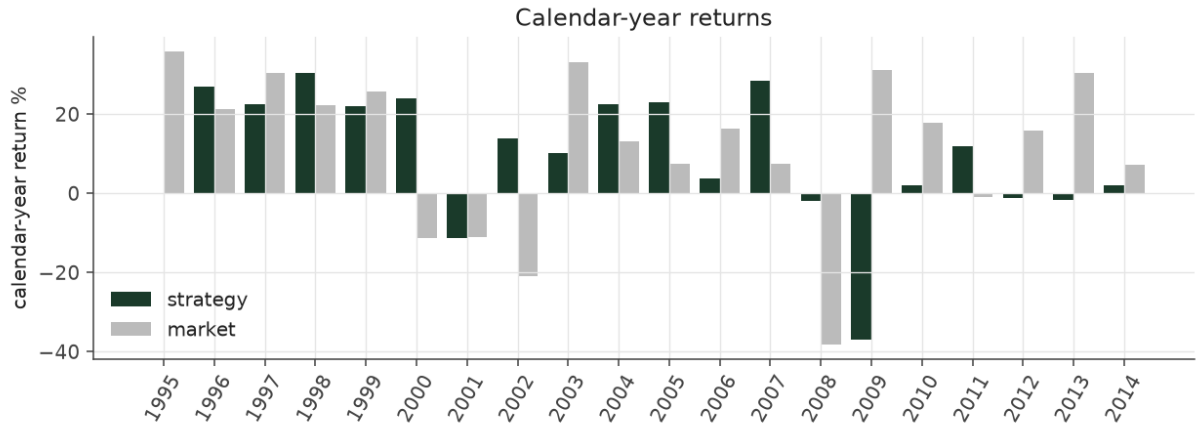
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.186



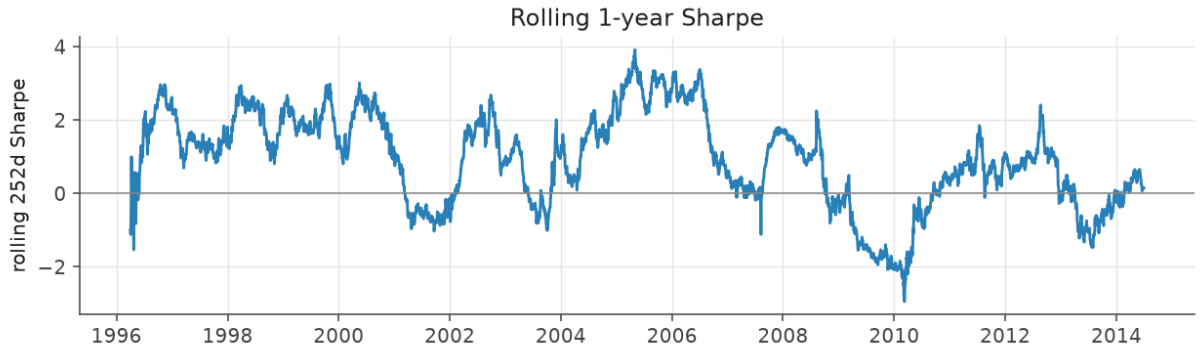
NAV (log scale)



Drawdown from running peak

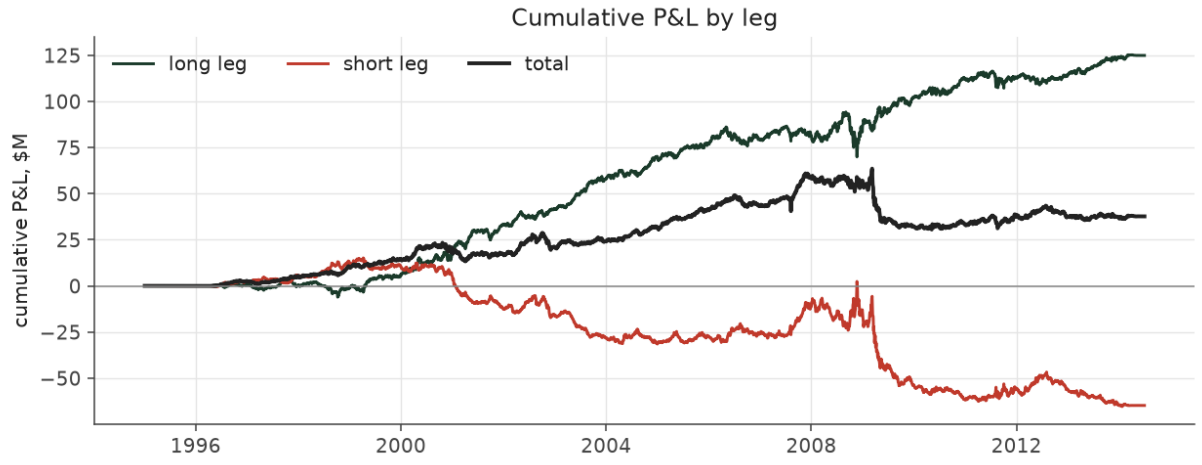


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	124.90	-64.83
P&L after costs ($M)	114.04	-76.37
of which common (average-event) component ($M)	119.86	-122.72

of which excess / PEAD component ($M)	5.05	57.89
excess minus costs ($M)	-5.82	46.35
gross return per $ traded (bp)	177.22	-88.04
positions in the money (hit rate)	0.55	0.48
stand-alone Sharpe (leg only)	1.00	-0.38

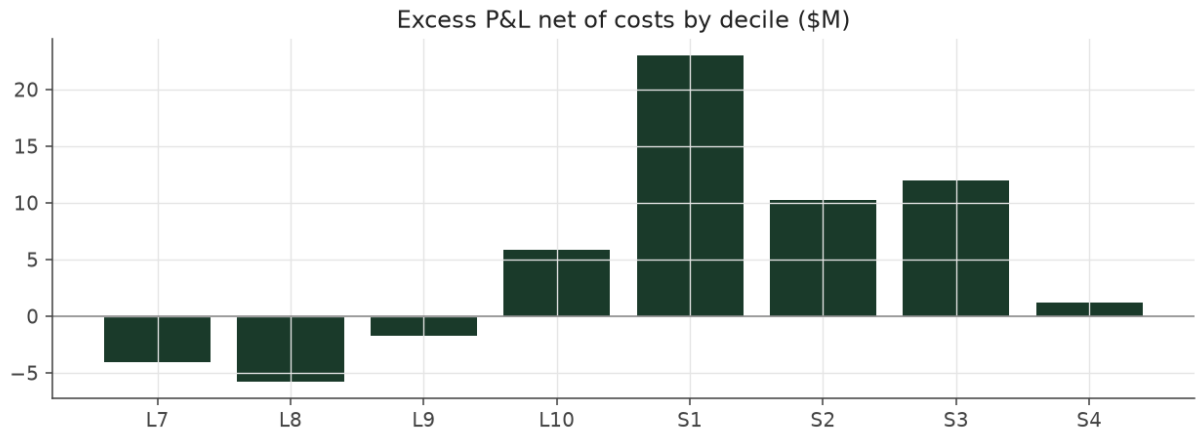
Daily P&L correlation between the two legs: **-0.74**.

• **The short leg loses money in absolute terms** (-64.8M before costs) while the long leg earns +124.9M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 119.9M / 122.7M. Stripping that common component, the surprise signal itself earned +5.0M on the long side and +57.9M on the short side.

• The two legs' daily P&L are -0.74 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.00 long / -0.38 short vs 0.69 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	3,244	418.72	3.15	-3.46	0.66	-4.11	-10.1%	-98.22
long	8	11,112	1,487.56	22.24	-3.49	2.32	-5.81	-14.3%	-39.07
long	9	13,857	2,195.43	32.84	1.63	3.34	-1.70	-4.2%	-7.76
long	10	16,409	2,946.33	55.81	10.36	4.55	5.81	14.3%	19.71
short	1	14,397	2,839.18	-24.90	27.52	4.54	22.98	56.7%	80.96
short	2	14,634	2,617.43	-34.07	14.28	4.07	10.22	25.2%	39.04
short	3	10,525	1,714.05	-15.26	14.62	2.65	11.97	29.5%	69.85
short	4	1,465	192.64	-2.14	1.46	0.28	1.18	2.9%	61.13



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	6,679	331.94	-5.30	-2.98	0.98	-3.97	-9.8%	-119.48
2	14,442	1,582.52	5.65	11.42	3.70	7.72	19.0%	48.77

3	19,426	3,099.11	12.52	18.91	5.81	13.10	32.3%	42.26
4	21,910	4,343.53	13.61	20.01	6.40	13.61	33.6%	31.32
5	23,186	5,054.24	11.20	15.59	5.51	10.08	24.9%	19.94

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	5,125	92.61	0.46	1.35	0.46	0.88	2.2%	95.45
2	14,300	1,181.02	-1.30	4.61	3.54	1.07	2.6%	9.03
3	20,824	3,382.54	18.20	27.98	6.76	21.21	52.3%	62.71
4	22,097	4,695.64	9.39	15.52	6.57	8.95	22.1%	19.05
5	23,293	5,058.34	10.80	13.34	5.06	8.28	20.4%	16.37

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	29,895	5,173.43	-11.36	33.59	7.78	25.80	63.7%	49.88
41-60d	20,346	3,298.60	5.39	21.14	5.64	15.49	38.2%	46.97
61-90d	29,276	5,201.79	34.05	0.54	7.44	-6.90	-17.0%	-13.27
>90d	6,126	737.53	9.58	7.68	1.54	6.14	15.1%	83.22

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	26,452	2,447.71	17.98	21.81	3.85	17.96	44.3%	73.39
2002-07	29,968	5,544.24	39.31	44.52	8.52	36.01	88.8%	64.94
2008-13	29,223	6,419.40	-19.62	-3.39	10.04	-13.44	-33.2%	-20.93

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	13,945	1,413.84	-5.32	-1.40	2.31	-3.71	-9.2%	-26.26
Chems	2,350	911.10	3.53	4.62	1.27	3.35	8.3%	36.80
Durbl	2,464	971.15	-1.63	0.69	1.37	-0.68	-1.7%	-6.99
Enrgy	5,523	1,239.10	10.99	13.53	1.79	11.73	29.0%	94.70
Hlth	8,778	1,529.54	8.27	10.27	2.44	7.83	19.3%	51.20
Manuf	9,286	1,371.31	1.37	3.28	2.21	1.07	2.6%	7.81
Money	9,738	1,006.41	3.77	5.27	1.76	3.51	8.7%	34.90
NoDur	4,402	1,165.92	5.68	6.90	1.73	5.17	12.8%	44.34
Other	11,612	1,284.04	6.67	9.27	2.18	7.09	17.5%	55.22
Shops	9,830	1,499.65	9.11	11.15	2.42	8.73	21.5%	58.20

Telcm	3,469	1,055.51	-4.45	-2.45	1.45	-3.90	-9.6%	-36.98
Utils	4,246	963.79	-0.32	1.80	1.46	0.34	0.8%	3.49

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	29,591	5,215.15	-10.61	1.87	7.58	-5.71	-14.1%	-10.95
low vol	27,167	4,306.69	25.27	33.04	7.18	25.86	63.8%	60.05
mid vol	27,433	4,712.18	21.91	26.96	7.34	19.63	48.4%	41.65

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	8.35%	0.00%
0.5%	7.1	7.46%	0.89%
1.0%	14.1	6.41%	1.94%
2.0%	28.3	3.46%	4.89%
5.0%	70.7	-100.00%	108.35%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.